

Mth642 Mid Term Subjective.

Short Question.

Question No. 1: Define the term Fluid?

Answer: A substance in the liquid or gas phase is referred to as a fluid.

Question No. 2: Define the term Stress?

Answer: **Stress** is defined as force per unit area and is determined by dividing the force by the area upon which it acts.

Question No. 3: Define the term Normal Stress?

Answer: The normal component of the acting on a surface per unit area is called the **normal stress**.

Question No. 4: Define the term Shear Stress?

Answer: The tangential component of force acting on a surface per unit area is called the **shear stress**.

Question No. 5: What is a Pressure?

Answer: In a fluid at rest, the normal stress is called **Pressure**.

Question No. 6: What is No-Slip Condition?

Answer: the flow of a fluid in a stationary pipe or over a solid surface that is nonporous solid “sticks” to the surface due to viscous effects and there is no slip. This is known as the **no slip condition**.

Question No. 7: Define the term Boundary Layer?

Answer: The flow region adjacent to the wall in which the viscous effects are significant is called the **boundary layer**.

Question No. 8: What is Flow Separation?

Answer: When a fluid is forced to flow over a curved surface, such as the back side of the cylinder at sufficiently high velocity the boundary layer can no longer remain attached to the surface and at same point it separates from the surface- a process called **flow separation**.

Question No. 9: Define the term Viscous Flows?

Answer: Flows in which the frictional effects are significant are called **viscous flows**.

Question No. 10: Define the term Incompressible Flow?

Answer: Incompressibility is an approximation, in which the flow is said to be incompressible if the density remains nearly constant throughout.

Question No. 11: What is Super Sonic and Sub Sonic?

Answer:

Super Sonic: A flow is called Super Sonic when Mach Number (**Ma**)>1.

Sub Sonic: A flow is called Sub Sonic when Mach Number (**Ma**)<1.

Question No. 12: Define the term Laminar Flow?

Answer: The highly ordered fluid motion characterized by smooth layers of fluid is called **laminar flow**.

Question No. 13: Define the term Steady and Un Steady Flow?

Answer:

Steady Flow:

The term steady implies no change of properties, velocity, temperature, etc., at a point with time.

Un Steady Flow:

The term Un Steady implies change of properties, velocity, temperature, etc., at a point with time.

Question No. 14: Write about Transient Term in N-S Equation?

Answer: Transient terms in N-S equations are zero.

$$\Rightarrow \partial u / \partial t = \partial p / \partial t = 0.$$

Question No. 15: Define the term Surrounding and Boundary?

Answer:

Surrounding: The mass or region outside the system is called the surroundings

Boundary: The real or imaginary surface that separates the system from its surroundings is called the boundary.

Question No. 16: Define the term Isolated System?

Answer: even energy is not allowed to cross the boundary, that system is called an isolated system.

Question No. 17: Write Full Form of USCS?

Answer: The full form of USCS is “United States Customary System”.

Question No. 18: Define the term Significant Digits?

Answer: Significant digits are digits that are relevant and meaningful. When performing calculations or manipulations of several parameters.

Question No. 19: Define the term Density and write it's formula and unit?

Answer: Density is defined as mass per unit volume.

Formula: $\rho = m/v$.

Unit: $\rho = \text{kg/m}^3$.

Question No. 20: Define the term Specific Gravity and write its formula?

Answer: Specific Gravity is defined as the ratio of the density of a substance to the density of some standard substance at a specified temperature

Question No. 21: Define the term Cavitation?

Answer: A phenomenon, which is a common cause for drop in performance and even the erosion of impeller blades, is called **cavitation**.

Question No. 22: Define the term Viscosity?

Answer: It appears that there is a property that represents the internal resistance of a fluid to motion or the “fluidity” and that property is the viscosity.

Question No. 23: Define the term Drag Force?

Answer: The force a flowing fluid exerts on a body in the flow direction is called the **drag force**.

Question No. 24: Write about Fluid Velocity in Steady Laminar Flow?

Answer: In steady laminar flow, the fluid velocity between the plates varies linearly between 0 and V, and thus the velocity profile and the velocity gradient are:

$\Rightarrow u(y) = (y/l)V$ and $du/dy = V/l$.

Where y is the vertical distance from lower plate.

Question No. 25: Write Angular Displacement or Deformation Formula?

Answer: The Formula is:

$\Rightarrow d\beta \approx \tan\beta = da/t = vdt/l = (du/dy)dt$

Question No. 26: Define the term Newtonian Fluids?

Answer: Fluids for which the rate of deformation is linearly proportional to the shear stress are called **Newtonian fluids**.

Question No. 27: Define the term Surface Tension and write its unit?

Answer: The magnitude of this force per unit length is called surface tension or coefficient of surface tension σ_s and is usually expressed in the unit N/m (or lbf/ft in English units).

Question No. 28: What is Meniscus?

Answer: The curved free surface of a liquid in a capillary tube is called the meniscus.

Question No. 29: Write Pressure Difference Formula?

Answer: The pressure difference between any two points 1 and 2 is:

$\Rightarrow \Delta P = P_2 - P_1 = -\int_1^2 \rho g dz$.

If. $P_1 = P_2$.

$\Rightarrow F_1/A_1 = F_2/A_2$.

$\Rightarrow F_1/F_2 = A_1/A_2$.

Question No. 30: What is Advective Acceleration?

Answer: The term $(V \cdot \nabla)V$ is called the advective acceleration and It accounts for the effect of the fluid particle moving to a new location in the flow.

Question No. 31: Define the term Material Derivative and write its formula?

Answer: The total derivative operator d/dt is called the **material derivative**.

Formula:

$\Rightarrow D/Dt = d/dt + (V \cdot \nabla)$.

Question No. 32: Write Full Form of CFD?

Answer: The full form of CFD is “**computational fluid dynamics**”.

Question No. 33: What is a Streamline and write it's formula?

Answer: A streamline is a curve that is everywhere tangent to the instantaneous local velocity vector.

Formula: In XY-Plane:

$\Rightarrow (dy/dx)_{\text{along a Streamline}} = v/u$.

Question No. 34: What is a Path lines?

Answer: A path line is the actual path traveled by an individual fluid particle over some time period.

Formula:

$\Rightarrow x = x_{(start)} + \int_{t_{(start)}}^1 V dt$.

Question No. 35: Define the term Linear Strain Rate and Shear Strain Rate?

Answer:

Linear Strain Rate: Linear strain rate is defined as the rate of increase in length per unit length.

Shear Strain Rate: Shear strain rate at a point is defined as half of the rate of decrease of the angle between two initially perpendicular lines that intersect at the point.

Question No. 36: Define the term Vorticity Vector?

Answer: The vorticity vector defined as the curl of the velocity vector V .

Formula:

$$\Rightarrow \xi \rightarrow = \nabla \rightarrow \times V \rightarrow = \text{Curl}(V \rightarrow)$$

Question No. 37: Write a Full Form of RTT?

Answer: The full of RTT is "Reynolds transport theorem".

Question No. 38: Write the formulas of Law of Conservation of Mass and Law of Conservation of Energy?

Answer:

Law of Conservation of Mass:

$$\Rightarrow m_{in} \dot{-} m_{out} \dot{-} = dm_{cv}/dt.$$

Law of Conservation of Energy:

$$\Rightarrow E_{int} \dot{-} E_{out} \dot{-} = dE_{cv}/dt.$$

Question No. 39: Define the term Bernoulli Equation?

Answer: An approximate relation between pressure, velocity, and elevation, and is valid in regions of steady, incompressible flow where net frictional forces are negligible.

Question No. 40: Define the term Stagnation Pressure?

Answer: The sum of the static and dynamic pressures is called the **stagnation pressure**. and it is expressed as

Formula:

$$\Rightarrow p_{stag} = P + \rho V^2/2.$$

Question No. 41: Write Navier Stokes Equation for Incompressible?

Answer: The Equation is:

$$\Rightarrow \rho D V \rightarrow / Dt = - \nabla \rightarrow P + \rho g \rightarrow + \mu \nabla^2 V \rightarrow$$

Question No. 42: Define Normalized Equation?

Answer: If the non-dimensional terms in the equation are of order unity, the equation is called **Normalized Equation**.

Question No. 43: State Divergence Theorem?

Answer: The quickest and most straight forward way to derive the differential form of conservation of mass is to apply the divergence theorem.

Divergence Theorem:

$$\Rightarrow \int_V u \rightarrow \cdot G \rightarrow dv = \int_A G \rightarrow \cdot n \rightarrow dA.$$

Question No. 44: Define the Mach Number and write its formula?

Answer: Ratio of the speed of flow to speed of sound is called Mach Number.

Formula: $Ma = V/C = \text{speed of flow} / \text{speed of sound}$.

Question No. 45: Why Light Year has dimension of Length?

Answer: The light year unit is then product of velocity and time.

Question No. 46: Name and Describe the types of motion of fluid particles?

Answer: The Name is:

- **Translation Motion:** A fluid particle moves from one location to another.
- **Rotation Motion:** A fluid particle rotates about an axis drawn through the particle.
- **Linear Strain Motion:** A fluid particle stretched in a direction such that a line segment in that direction is elongated at some later time.

Question No. 47: Define the term Internal Flow with example?

Answer: Internal Flow is defined as a flow where in the fluid is completely confined by inner surface of an item.

Example: Flows through pipes ducts, nozzles and fittings.

Question No. 48: Write difference between Viscous and Inviscid Flow?

Answer:

Viscous Flow: The flows in which frictional effects are significant are called **viscous flow**.

Inviscid Flow: **Inviscid Flow** is the flow of an Inviscid fluid in which the viscosity of fluid is equal to zero.

Question No. 49: Write formula for finding Irrotational Flow?

Answer: The Formula is:

$$\Rightarrow \frac{1}{2}(\delta v / \delta x - \delta u / \delta y) = 0.$$

Question No. 50: Write formula for finding Incompressible Flow?

Answer: The Formula is:

$$\Rightarrow \delta u / \delta x + \delta v / \delta y = 0.$$

Question No. 51: Write Application of Fluid Mechanics in Medicine?

Answer: Fluid Mechanics is also going through your body for example Chemotherapy and Blood Thinner etc.

Question No. 52: Write Rate of Translation Vector in Cartesian Coordinates?

Answer: The Formula of Rate of Translation Vector in Cartesian Coordinates is:

$$\Rightarrow \vec{V} = u\hat{i} + v\hat{j} + w\hat{k}.$$

Question No. 53: Write Rate of Rotation Vector in Cartesian Coordinates?

Answer: The Formula of Rate of Rotation Vector in Cartesian Coordinates is:

$$\vec{W} = \frac{1}{2}(\delta w / \delta y - \delta v / \delta z)\hat{i} + \frac{1}{2}(\delta u / \delta z - \delta w / \delta x)\hat{j} + \frac{1}{2}(\delta v / \delta x - \delta u / \delta y)\hat{k}.$$

Question No. 54: Write Difference Between Solid and Fluid?

Answer:

Solid: The Shape and Volume of Solid is Definite. Solid have high density. Solid can't compressed.

Fluid: Fluid is a liquid then The shape of fluid is not definite but its volume is definite. Fluid have low density. Fluid can be compressed.

Question No. 55: Define the Ideal Fluid?

Answer: A fluid is called Ideal Fluid if it has following Characters.

- The fluid is non viscous.
- The fluid is Incompressible.
- The fluid motion is Steady.

Question No. 56: Define the term Fluid kinematics?

Answer: Fluid Kinematics deals with the motion of fluids without considering the forces and moments which create the motion.

Question No. 57: Define the term Mass Flow Rate and Write its formula?

Answer: The amount of mass flowing through a cross section per unit time.

Formula: $\delta m^* = \rho V_n dA_c$

Long Question:

Question No. 1: Write Application Areas of Fluid Mechanics?

Answer: The application areas of Fluid Mechanics is:

- Fluid mechanics is widely used both in everyday activities and in the design of Modern engineering system.
- Fluid mechanics plays a vital role in the human body.
- All artificial hearts, breathing machines and dialysis systems are designed Using fluid dynamics.
- Fluid mechanics plays a major role in the design of all these components.
- We can also see numerous applications in an automobile.
- The piping systems of cold water, natural gas and sewage for an individual House are designed primarily on the basis of fluid Mechanics.
- Fluid mechanics is also used in the design of the heating and air-conditioning System
- The cooling system of the engine block including the radiator and the water Pump, and even the tires.

Question No. 2: Write Internal and External Regions of Flow?

Answer:

- A fluid flow is classified as being internal or external, depending on whether the fluid flows in a confined space or over a surface.
- The flow of an unbounded fluid over a surface such as a plate, a wire, or a pipe is external flow.
- The flow in a pipe or duct is internal flow if the fluid is bounded by solid surfaces.
- Water flow in a pipe, for example, is internal flow, and airflow over a ball or over an exposed pipe during a windy day is external flow.
- The flow of liquids in a duct is called open-channel flow if the duct is only partially filled with the liquid and there is a free surface.
- The flows of water in rivers and irrigation ditches are examples of such flows.
- Internal flows are dominated by the influence of viscosity throughout the flow field.
- In external flows the viscous effects are limited to boundary layers near solid surfaces and to wake regions downstream of bodies.

Question No. 3: Difference Between Accuracy Error and Precision Error?

Answer: The Difference Between Accuracy Error and Precision Error:

Accuracy Error: Accuracy error (inaccuracy) is the value of one reading minus the true value.

Closeness of the average reading to the true value. Generally associated with repeatable, fixed errors.

Precision Error: Precision error is the value of one reading minus the average of readings. A measure of the fineness of the resolution and the repeatability of them instrument. Generally associated with unrepeatable, random errors.

Question No. 4: Write uses of Pascal Law?

Answer: The Uses of Pascal Law:

- A consequence of the pressure in a fluid remaining constant in the horizontal direction is that the pressure applied to a confined fluid increases the pressure throughout by the same amount. This is called Pascal's law, after Blaise Pascal (1623–1662).
- Pascal also discovered that the force applied by a fluid is proportional to the surface area. He realized that two hydraulic cylinders of different areas could be connected, and the larger could be used to exert a proportionally greater Force than that applied to the smaller.
- "Pascal's machine" has been the source of many inventions that are a part of our daily lives such as hydraulic brakes and lifts. This is what enables us to lift a car easily by one arm.

Question No. 5: Write Acceleration Field in Vector Form?**Answer:****In Vector Form:**

$$\Rightarrow a_{(\text{particle})}^{\rightarrow}(x, y, z, t) = dV^{\rightarrow}/dt.$$

$$\Rightarrow a_{(\text{Particle})}^{\rightarrow}(x, y, z, t) = \partial V^{\rightarrow}/\partial t + (\nabla^{\rightarrow} \cdot \nabla^{\rightarrow})V^{\rightarrow}.$$

The Gradient or del operator:

$$\Rightarrow \nabla^{\rightarrow} = (\partial/\partial x, \partial/\partial y, \partial/\partial z) = \partial/\partial x i^{\wedge} + \partial/\partial y j^{\wedge} + \partial/\partial z k^{\wedge}.$$

And In Cartesian Coordinate:

$$\Rightarrow a_x = \partial u/\partial t + u\partial u/\partial x + v\partial u/\partial y + w\partial u/\partial z.$$

$$\Rightarrow a_y = \partial v/\partial t + u\partial v/\partial x + v\partial v/\partial y + w\partial v/\partial z.$$

$$\Rightarrow a_z = \partial w/\partial t + u\partial w/\partial x + v\partial w/\partial y + w\partial w/\partial z.$$

Then:

$$\Rightarrow a_{(\text{particle})}^{\rightarrow}(x, y, z, t) = dV^{\rightarrow}/dt = \partial V^{\rightarrow}/\partial t + (\nabla^{\rightarrow} \cdot \nabla^{\rightarrow})V^{\rightarrow}.$$

Question No. 6: Define Manometer and Write its uses?**Answer:**

- We notice from equation:

$$\Rightarrow \Delta P = P_2 - P_1 = -\rho g \Delta z = -\gamma_s \Delta z = -\int_1^2 \rho g dz.$$

That an elevation change of $-\Delta z$ in a fluid at rest corresponds to $\Delta P/\rho g$, which suggests that a fluid column can be used to measure pressure differences. A device based on this principle is called a manometer, and it is commonly used to measure small and moderate pressure differences.

- A manometer consists of a glass or plastic U-tube containing one or more fluids such as mercury, water, alcohol, or oil.
- To keep the size of the manometer to a manageable level, heavy fluids such as mercury are used if large pressure differences are anticipated.

Question No. 7: Define the term Linear Strain Rate and Shear Strain Rate with formulas?**Answer:****Linear Strain Rate:**

Linear strain rate is defined as the rate of increase in length per unit length.

In Cartesian Coordinates:

$$\Rightarrow \epsilon_{xx} = \partial u/\partial x, \epsilon_{yy} = \partial v/\partial y, \epsilon_{zz} = \partial w/\partial z.$$

Shear Strain Rate:

Shear strain rate at a point is defined as half of the rate of decrease of the angle between two initially perpendicular lines that intersect at the point.

In Cartesian Coordinates:

$$\Rightarrow \epsilon_{xy} = 1/2(\partial u/\partial y + \partial v/\partial x),$$

$$\Rightarrow \epsilon_{zx} = 1/2(\partial w/\partial x + \partial u/\partial z),$$

$$\Rightarrow \epsilon_{yz} = 1/2(\partial v/\partial z + \partial w/\partial y).$$

Question No. 8: Define the Rate of Translation and Rate of Rotation with Formulas?

Answer:

Rate of Translation:

The Formula is:

$$\Rightarrow \vec{V} = u\hat{i} + v\hat{j} + w\hat{k}.$$

Rate of Rotation:

Rate of rotation (angular velocity) at a point is defined as the average rotation rate of two initially perpendicular lines that intersect at that point.

Formula:

$$\Rightarrow \vec{\omega} = 1/2(\delta w/\delta y - \delta v/\delta z)\hat{i} + 1/2(\delta u/\delta z - \delta w/\delta x)\hat{j} + 1/2(\delta v/\delta x - \delta u/\delta y)\hat{k}.$$

Question No. 9: Write Comparison of Two Circular Flows?

Answer:

All Flows with Circular Streamline are not rotational.

Flow A-Solid Body Rotation:

$$\Rightarrow u_r = 0 \text{ and } u_\theta = \omega r.$$

Flow B-Line Vertex:

$$\Rightarrow u_r = 0 \text{ and } u_\theta = k/r.$$

Flow A-Solid Body Rotation:

$$\Rightarrow \vec{\xi} = 1/r(\partial(\omega r^2)/\partial r - 0)\hat{k}.$$

$$\Rightarrow \vec{\xi} = 2\omega\hat{k}.$$

Flow B-Line Vertex:

$$\Rightarrow \vec{\xi} = 1/r(\partial(k)/\partial r - 0)\hat{k} = 0.$$

Question No. 10: Reduce the following expression as far as possible.

$$F(t) = d/dt \int_{x=At}^{x=Bt} e^{-2x^2} dx.$$

Answer:

F(t) is to be evaluated from the given expression:

The integral is:

$$F(t) = d/dt \int_{x=At}^{x=Bt} e^{-2x^2} dx.$$

Here $G(x, t) = e^{-2x^2}$. and the limits of integration are $a(t) = At$ and $b(t) = Bt$. **Thus:**

$$\Rightarrow F(t) = \int_a^b \partial G/\partial t dx + db/dt G(b, t) - da/dt G(a, t).$$

$$\Rightarrow F(t) = 0 + Be^{-2b^2} - Ae^{-2a^2}.$$

$$\Rightarrow F(t) = Be^{-2b^2} - Ae^{-2a^2}.$$

Question No. 11: Write Three Dimensional Leibniz Theorem?

Answer:

Three Dimensional Leibniz Theorem:

$$\Rightarrow d/dt \int_{V(t)} G(x, y, z, t) dV = \int_{V(t)} \partial G/\partial t dV + \int_{A(t)} G \vec{V}_A \cdot \vec{n} dA.$$

where $v(t)$ is a moving volume, $A(t)$ is its Surface and $\vec{V}_A$ is absolute velocity of this surface.

$$\Rightarrow d/dt \int_{V(t)} \rho b dV = \int_{V(t)} \partial(\rho b)/\partial t dV + \int_{A(t)} \rho b \mathbf{V}_A \cdot \mathbf{n} dA.$$

If we apply Leibniz Theorem to the special case of a material volume. Then $\mathbf{V}_A = \mathbf{V}$.

Thus:

$$\Rightarrow d/dt \int_{V(t)} \rho b dV = dB_{sys}/dt = \int_{V(t)} \partial(\rho b)/\partial t dV + \int_{A(t)} \rho b \mathbf{V} \cdot \mathbf{n} dA.$$

Question No. 12: Write Derivation of Bernoulli Equation?

Answer:

- The forces acting on a fluid particle along a streamline.
- The sum of the kinetic, potential, and flow energies of a fluid particle is constant along a streamline during steady flow when compressibility and frictional effects are negligible.

$$\Rightarrow P dA - (P + dP) dA - W \sin \theta = m V dV / dS \dots \dots \dots \text{eq(1)}.$$

Where:

- $M = \rho V = \rho dA dS.$
- $W = mg = \rho g dA dS.$
- $\sin \theta = dZ / dS.$

Substituting equation(1).

$$\Rightarrow -dP dA - \rho g dA dS dZ / dS = \rho dA S V dV / dS.$$

$$\Rightarrow -dP - \rho g dZ = \rho V dV.$$

Put $V dV = 1/2 d(V^2).$

$$\Rightarrow dP / \rho + 1/2 d(V^2) + g dZ = 0.$$

Integrating, we have:

$$\Rightarrow \int (dP / \rho + 1/2 d(V^2) + g dZ) = 0.$$

Question No. 13: If $\mathbf{V} = x^2 + 2x - y$, then find flow is rotational or irrotational?

Answer:

As we know:

$$\Delta \times \mathbf{V} = 0 \dots \dots \dots \text{eq(1)}.$$

If $\Delta \times \mathbf{V} = 0$ then it will irrotational.

If $\Delta \times \mathbf{V} \neq 0$ then it will rotational.

In Order to find curl of $\mathbf{V}$.

first we have to do: $\mathbf{V} = \Delta \phi.$

Here we will put value of $\mathbf{V}$ and $\Delta \phi.$

$$\Rightarrow x^2 + 2x - y = \delta \phi / \delta x \mathbf{i} + \delta \phi / \delta y \mathbf{j} + \delta \phi / \delta z \mathbf{k}.$$

$$\Rightarrow x^2 + 2x - y = (2x + 2) \mathbf{i} - \mathbf{j} + 0 \mathbf{k}.$$

Now put value in eq(1).

$$\Delta \times \mathbf{V} = 0.$$

$$\Rightarrow \Delta \times \mathbf{V} = \mathbf{i} (0 - 0) - \mathbf{j} (0 - 0) + \mathbf{k} (0 - 0).$$

$$\Rightarrow \Delta \times \mathbf{V} = 0.$$

So, it is irrotational.

Question No. 14: Write Linear Momentum Equation in Newton's Second Law?

Answer:

Newton's Second Law:

$$\Rightarrow F = ma.$$

where F is force acting on an object, m is mass of object and a is the objects acceleration.

We can rewrite this using the definition $a = \Delta v / \Delta t$, where Δv is change in velocity of the object over some time duration Δt .

$$\Rightarrow F = m \Delta v / \Delta t.$$

which is equivalent to:

$$\Rightarrow F = \Delta p / \Delta t.$$

where $\Delta p = m \Delta v$ is change in momentum.

Question No. 15: Write Compressible and Incompressible Continuity Equation?

Answer:

Compressible Flow:

$$\Rightarrow \nabla \cdot (\rho \mathbf{V}) = 0.$$

Compressible Continuity Equation:

In Cartesian Coordinates:

$$\Rightarrow \partial(\rho u) / \partial x + \partial(\rho v) / \partial y + \partial(\rho w) / \partial z = 0.$$

In Cylindrical Coordinates:

$$\Rightarrow \partial(r \rho u_r) / r \partial r + \partial(\rho u_\theta) / r \partial \theta + \partial(\rho u_z) / \partial z = 0.$$

And:

Incompressible Flow:

$$\Rightarrow \nabla \cdot \mathbf{V} = 0.$$

Since $\partial \rho / \partial t = 0$.

Incompressible Continuity Equation:

In Cartesian Coordinates:

$$\Rightarrow \partial u / \partial x + \partial v / \partial y + \partial w / \partial z = 0.$$

In Cylindrical Coordinates:

$$\Rightarrow \partial(r u_r) / r \partial r + \partial(u_\theta) / r \partial \theta + \partial u_z / \partial z = 0$$

The End.